

Why Do States Abandon the Inheritance Tax?

The Role of Revenue Efficiency and Democratic Institutions

Applied Statistical Analysis II

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1 Original Study

I replicate the following study: "Genschel, P., Limberg, J., & Seelkopf, L. (2024). Revenue, Redistribution, and the Rise and Fall of Inheritance Taxation. *Comparative Political Studies*, 57(9), 1475-1505."

1.1 Background and Research Question

- **Research Question:** Why do many countries repeal the inheritance tax after having previously introduced it?
- **Limitations of Existing Literature:** Conventional public policy research suggests that once policies are implemented, they tend to persist due to institutional inertia and entrenched interests. However, inheritance taxation represents a notable exception, having experienced a widespread decline since the mid-20th century.
- **Authors' Perspective:** The authors argue that the "rise and fall" of the inheritance tax is better understood through changes in its fiscal role within the state. In particular, they emphasize that its trajectory is primarily driven by transformations in its revenue function, rather than by purely political or ideological factors such as democratic pressure or elite capture.

1.2 Hypotheses

- **Revenue Hypothesis:** The likelihood of inheritance tax repeal increases as governments adopt and expand more efficient and stable revenue instruments, such as income taxes and Value Added Tax (VAT). As these alternative sources grow, the fiscal necessity of inheritance taxation declines, making it more politically expendable.
- **Redistribution Hypothesis:** Once the fiscal importance of inheritance taxation weakens, its survival depends on redistributive considerations. In this context, democratic regimes—where redistributive demands are more politically salient—are expected to be less likely to repeal the tax than non-democratic regimes.

1.3 Data and Method

- **Data:** The authors use a novel dataset tracking inheritance tax introductions and repeals for a global sample of 87 countries over approximately two centuries (1759–2015).
- **Method:** The study utilizes a Binary Time-Series Cross-Section (BTSCS) dataset, where country-years are observed until the repeal of the inheritance tax.
- **Model:** The analysis is conducted using logistic regression models estimated via maximum likelihood. To account for temporal dependence in BTSCS data, a cubic polynomial approximation (t, t^2, t^3) is included to model time at risk.

Figure 1: Inheritance Tax Introductions and Repeals over Time

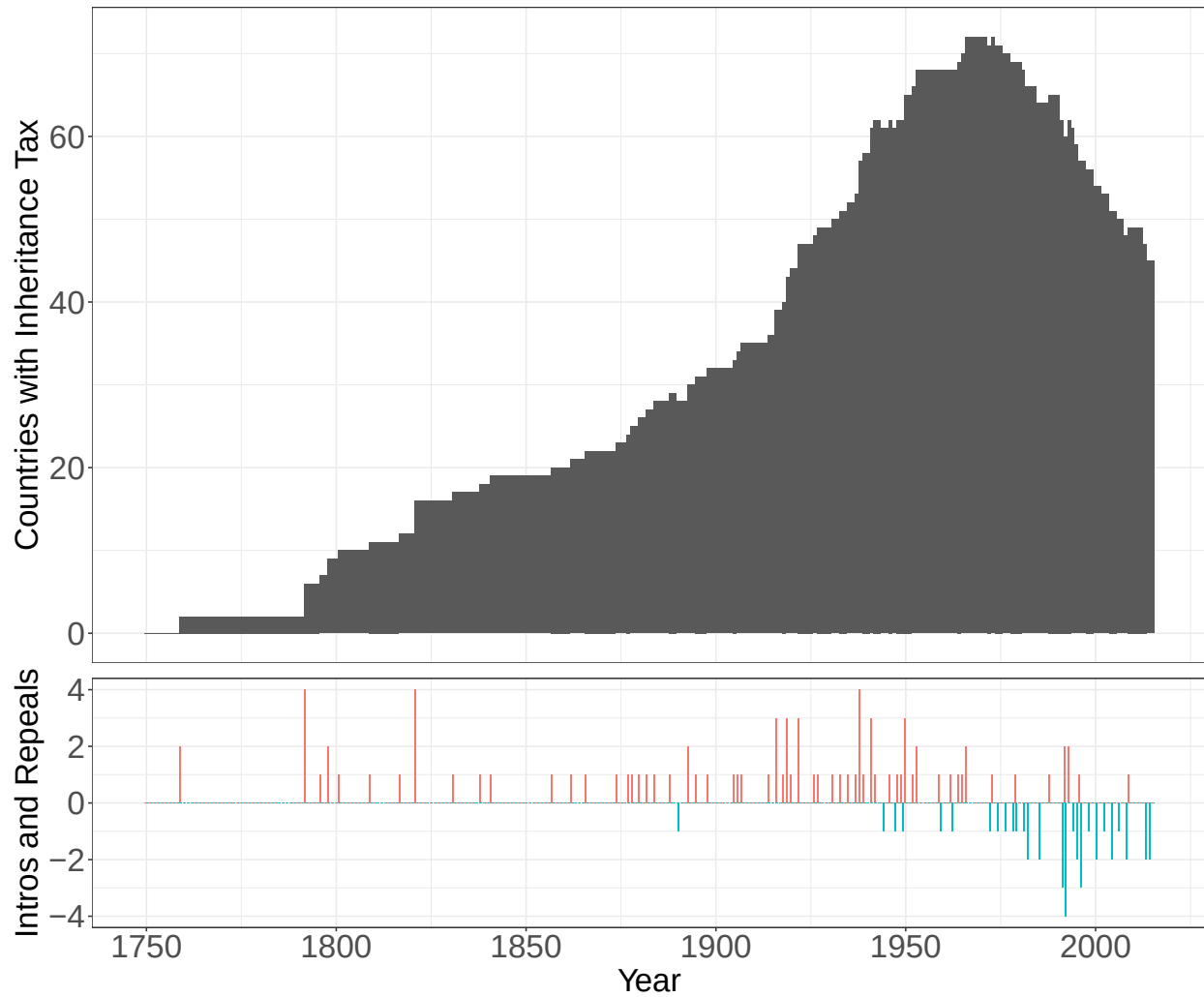


Figure 1 Description: The number of countries levying an inheritance tax increased steeply during the 19th and early 20th centuries but has declined rapidly since the 1960s as repeals began to outpace introductions.

2 Data

The main analysis is conducted using the `merge_full_2` dataset, which is a Binary Time-Series Cross-Section (BTSCS) dataset tracking countries at risk of repealing their inheritance tax over time.

2.1 Outcome Variable

Inheritance tax repeal (y), coded as:

- Inheritance tax is permanently repealed: 1
- Inheritance tax is maintained (or censored): 0

2.2 Input Variables

2.2.1 Key Independent Variables:

- **Major modern taxes (`other_tax`):** An index measuring the introduction of income and consumption taxes, coded as 0 (neither), 1 (either), or 2 (both).
- **Democracy (`democracy`):** A binary indicator of democratic regime type.

2.2.2 Control Variables:

- **Fiscal/Economic:** Includes indicators for recent war (`war_previous`), recent economic recession (`recession_prev_5`), and social policy introduction (`wfs_intro_previous`).
- **Demographic/Structural:** Includes logged population for tax competition (`log(population_new)`), logged GDP per capita (`log_gdp_cap`), life expectancy (`e_pelifeex`), and logged time since the tax was introduced (`log(time_since_intro)`).
- **Temporal Dependence (t, t_2, t_3):** A cubic polynomial approximation of time at risk to account for temporal dependence in the BTSCS data.

3 Original Model

The authors build their argument by first exploring the historical conditions of the inheritance tax's introduction and consolidation, followed by a logistic regression model to analyze the determinants of its repeal.

3.1 Historical Context (Figures 2, 3, and 4)

Before testing the repeal of the inheritance tax, the authors illustrate why it was introduced in the first place. Figure 2 shows that introductions were closely associated with spending-intensive events like wars, recessions, and the introduction of social policies. Figure 3 demonstrates that inheritance taxes were mostly adopted before other modern taxes (income and consumption taxes), highlighting their crucial role when alternative revenues were scarce. The R code snippet to replicate Figure 3 is as follows:

```
1 fig3 <- rank_inheritance %>%
2   ggplot(., aes(x=rank_tax_base, fill =rank_tax_base )) +
3   geom_bar(aes(y = after_stat(count)/sum(after_stat(count)))) +
4   scale_y_continuous(labels=scales::percent) +
5   theme_bw() +
6   ylab("Share of Countries") +
7   scale_fill_grey() +
8   xlab("Historical Timing Inheritance Tax") +
9   theme(text = element_text(size=20),
10  legend.position = "none",
11  axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))
```

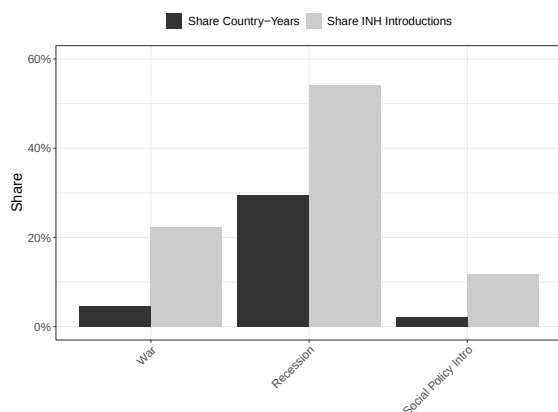


Figure 2: Context of Introductions

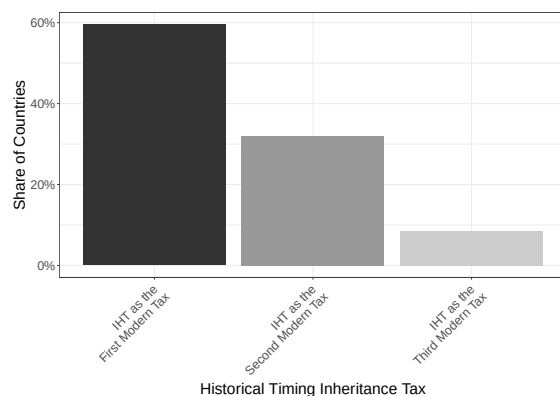
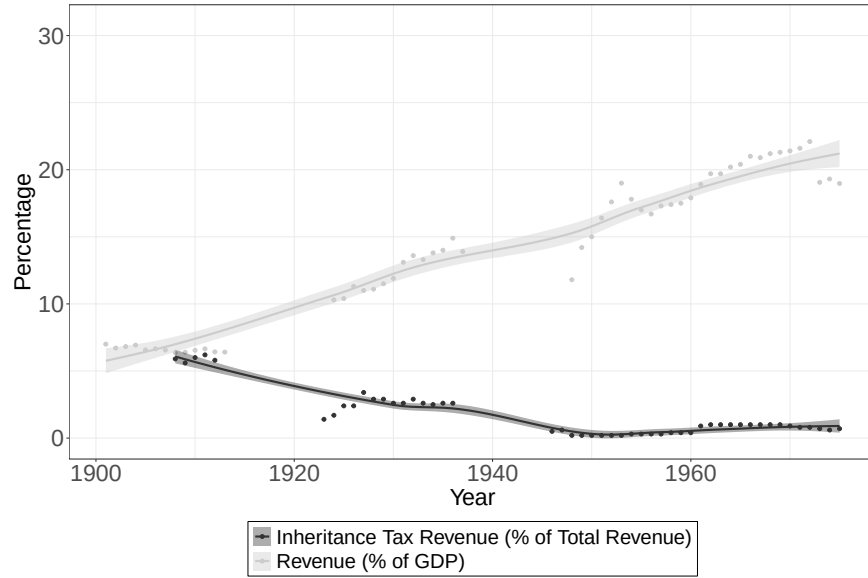


Figure 3: Timing of IHT

However, as other efficient revenue instruments were developed, the fiscal significance of the inheritance tax declined. Figure 4 uses Austria as a case study to show how the inheritance tax revenue as a percentage of total revenue steadily dropped over the 20th century.

Figure 4:



3.2 Main Regression Model (Table 1)

To test their hypotheses on tax repeal, the authors estimate a series of logistic regression models using maximum likelihood estimation. To account for temporal dependence in the Binary Time-Series Cross-Section (BTSCS) data, they include a cubic polynomial approximation (t , t^2 , t^3).

The R code to replicate the fully specified model (Model 3) is as follows:

```
1 main_inh_3 <- glm(y ~ other_tax + democracy +  
2 war_previous + recession_prev_5 + wfs_intro_previous +  
3 log(time_since_intro) + log_gdp_cap + e_pelifeex + log(population_new)  
+ t + t2 + t3,  
4 data = merge_full_2, family=binomial(link="logit"))
```


Table 1: Regression Results for Inheritance Tax Repeal

	Model 1	Model 2	Model 3
Major Modern Taxes	1.576*** (0.418)	1.546*** (0.417)	1.098** (0.453)
Democracy	-0.779** (0.356)	-0.763** (0.359)	-1.197*** (0.387)
War		0.199 (1.032)	0.764 (1.066)
Recession		0.266 (0.355)	0.384 (0.356)
Social Policy Intro		-15.602 (570.644)	-15.311 (568.308)
Tax Competition (Population log)			0.034 (0.127)
Time Since Intro (log)			-0.415 (0.348)
Life Expectancy			0.062** (0.025)
GDP pc (log)			-0.117 (0.277)
t	0.052 (0.041)	0.054 (0.041)	0.080 (0.052)
t2	-0.001 (0.001)	-0.001 (0.001)	-0.001* (0.001)
t3	0.000* (0.000)	0.000* (0.000)	0.000* (0.000)
AIC	464.157	457.395	452.416
BIC	504.515	517.931	539.697
Log Likelihood	-226.079	-219.697	-213.208
Deviance	452.157	439.395	426.416
Num. obs.	6163	6163	6087

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

The results indicate that the coefficient for **Major Modern Taxes** is positive and statistically significant, suggesting that the expansion of alternative revenue sources increases the likelihood of inheritance tax repeal. Substantively, this supports the argument that inheritance taxation becomes politically vulnerable once it is no longer fiscally indispensable.

In contrast, the coefficient for **Democracy** is negative and statistically significant, indicating that democratic regimes are less likely to repeal inheritance taxes. This finding aligns with the redistribution hypothesis, suggesting that democratic institutions may sustain redistributive policies even when their fiscal importance declines.

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Overall, these results provide strong empirical support for the authors' central claim that fiscal structure, rather than purely political ideology, plays a decisive role in shaping the fate of inheritance taxation.

The replicated estimates closely match those reported in the original study in terms of coefficient signs and statistical significance. Minor differences may arise from data preprocessing or sample restrictions, but the substantive conclusions remain unchanged, indicating that the original findings are robust and reproducible.

3.3 Robustness Checks (Figures 5 and 6)

Furthermore, the authors present dot-and-whisker plots (Figures 5 and 6) to demonstrate the robustness of their main findings using alternative measurements for revenue capacity and democracy. The code to replicate Figure 6 is found in the attached R file. I provide a snippet below:

```
1 # Visualizing Robustness Checks (Coefficient Plots)
2 ggplot(all_treat, aes(x = fct_rev(model), y = estimate)) +
3   geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0,
4     linewidth = 1.1) +
5   theme_bw() + coord_flip() +
6   geom_hline(yintercept = 0, linetype = "dashed") +
7   geom_point(size = 2) +
8   labs(y = "Point Estimates and 90% CI (Standardised)", x = "")
```

Figure 5: Robustness Check: Revenue Capacity

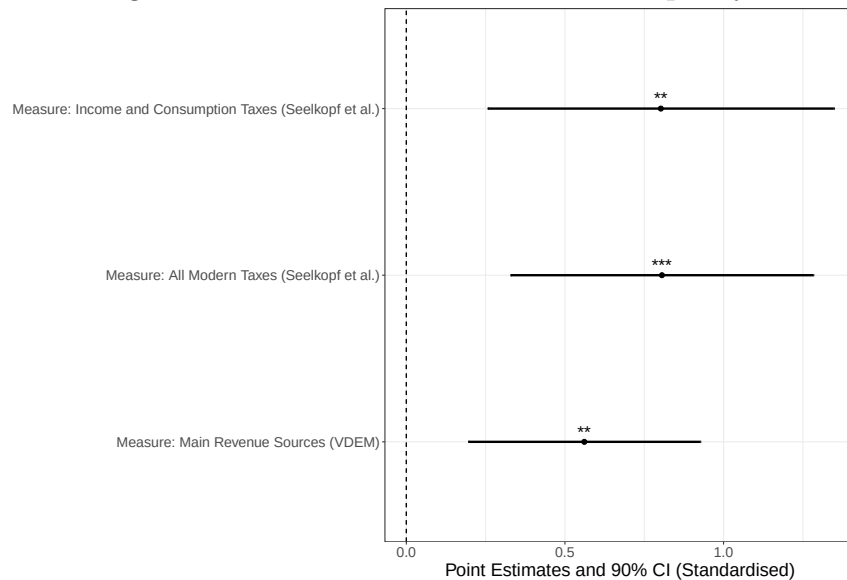
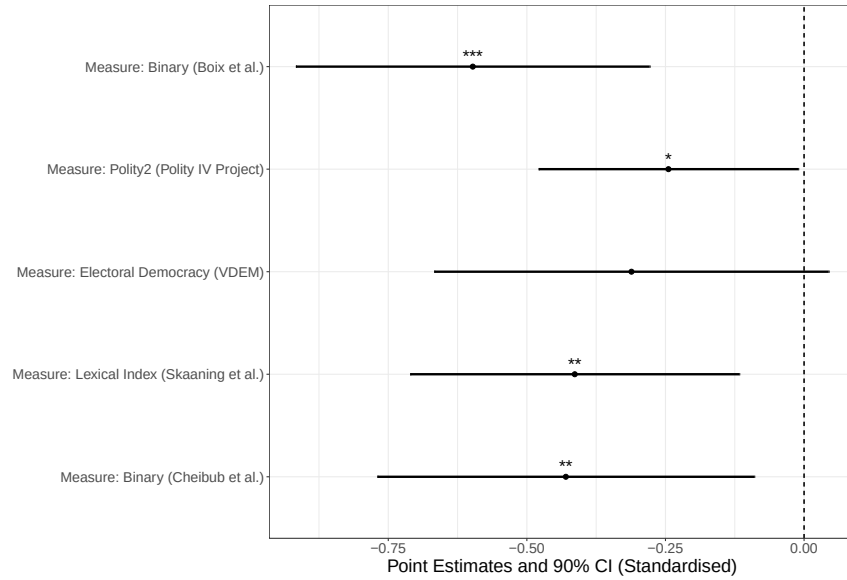


Figure 6: Robustness Check: Democracy Measures



According to the authors, the coefficients in these models demonstrate that "the likelihood of inheritance tax repeal increases as governments adopt and expand more efficient revenue instruments," while "democracies are less likely to repeal the inheritance tax than autocracies."

4 My Contribution

While the original model provides valuable insights, the interpretation of log-odds coefficients remains substantively unintuitive. For instance, the coefficient for **Major modern taxes** (1.098) indicates that a one-unit increase in the index raises the log-odds of repeal, holding other variables constant. While this conveys direction and statistical significance, it does not provide an intuitive sense of how the probability of repeal changes in practice.

Moreover, the original model assumes that the effects of fiscal capacity and democracy are strictly additive. This may be overly restrictive, as democratic institutions could plausibly moderate the relationship between revenue conditions and policy outcomes.

Therefore, in this section, I extend the original model by incorporating an interaction term and translating the results into predicted probabilities, providing a more intuitive and theoretically nuanced interpretation.

4.1 The Interaction Hypothesis

I argue that we can formulate a theoretical interaction hypothesis:

Hypothesis 1. *The impact of modern taxes on the likelihood of inheritance tax repeal varies depending on the level of democracy.*

According to the authors, democracies resist inheritance tax repeals because they protect the redistributive interests of the lower classes. If this is true, even when a state achieves “revenue redundancy” by adopting efficient modern taxes, democratic institutions should resist the fiscal pressure to repeal the inheritance tax more strongly than autocracies. Therefore, the positive effect of the **other_tax** variable should be weaker in democratic regimes.

4.2 Statistical Concept Application and Visualization

To test this hypothesis, I apply the concepts learned in Applied Stats II by adding an interaction term (**other_tax * democracy**) to the original logistic regression model. Furthermore, to move beyond the unintuitive log-odds interpretation, I use the **marginalEffects** package in R. By plotting the predicted probability S-curves, I visually demonstrate how the probability of an inheritance tax repeal behaves differently for democracies versus autocracies as the modern tax index increases.

4.2.1 Table 2: Interaction Model

```
1 # Fit interaction model: other_tax * democracy
2 model_int <- glm(y ~ other_tax * democracy + war_previous +
3 recession_prev_5 + wfs_intro_previous +
4 log(time_since_intro) + log_gdp_cap + e_pelifeex +
5 log(population_new) + t + t2 + t3,
6 data = merge_full_2, family = binomial(link = "logit"))
7
```

Table 2: Determinants of Inheritance Tax Repeals (Interaction Model)

	Original (Model 3)	Interaction Model
Major Modern Taxes	1.098** (0.453)	1.227** (0.548)
Democracy	-1.197*** (0.387)	-0.429 (1.691)
Major Modern Taxes $\times$ Democracy		-0.417 (0.899)
War	0.764 (1.066)	0.756 (1.066)
Recession	0.384 (0.356)	0.382 (0.356)
Social Policy Intro	-15.311 (568.308)	-15.318 (569.734)
Tax Competition (Population log)	0.034 (0.127)	0.028 (0.127)
Time Since Intro (log)	-0.415 (0.348)	-0.430 (0.350)
Life Expectancy	0.062** (0.025)	0.061** (0.025)
GDP pc (log)	-0.117 (0.277)	-0.111 (0.277)
t^1	0.080 (0.052)	0.081 (0.053)
t^2	-0.001* (0.001)	-0.001* (0.001)
t^3	0.000* (0.000)	0.000* (0.000)
AIC	452.416	454.211
BIC	539.697	548.205
Log Likelihood	-213.208	-213.105
Deviance	426.416	426.211
Num. obs.	6087	6087

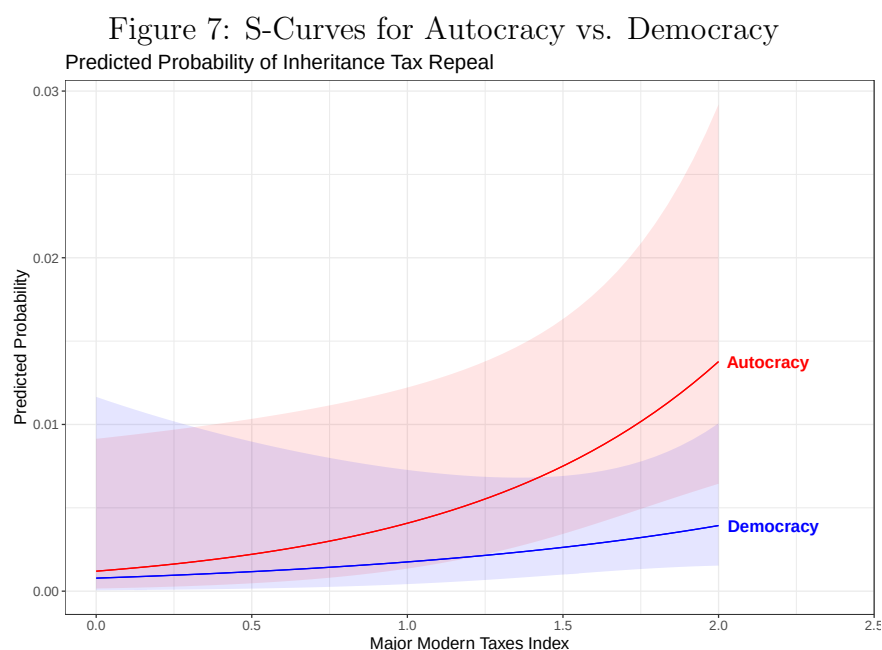
*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Note: Table 2 shows that the interaction term Major Modern Taxes $\times$ Democracy is negative (-0.417), aligning with the theoretical expectation that democratic institutions attenuate the effect of revenue redundancy. However, the interaction term does not achieve statistical significance at conventional levels, suggesting that the empirical evidence for this moderating effect remains inconclusive.

4.2.2 Figure 7: Predicted Probability S-Curves

As visualized in Figure 7, when the modern tax index increases, the predicted probability of an inheritance tax repeal rises sharply in autocracies (red line). In contrast, in democratic regimes (blue line), this curve is significantly flattened, remaining below a 0.01 probability even at the highest levels of the modern tax index.

```
1 # Generate Predicted Probability S-Curves
2 fig7 <- plot_predictions(model_int, condition = c("other_tax", "
democracy")) +
3 theme_bw() +
4 labs(title = "Predicted Probability of Inheritance Tax Repeal",
5 x = "Major Modern Taxes Index", y = "Predicted Probability") +
6 scale_color_manual(values = c("red", "blue"), labels = c("Autocracy",
"Democracy"))
7
```



Although the interaction term itself is not statistically significant—likely due to the overall rarity of repeal events (only 40 out of 6,087 observations)—the graphical representation provides suggestive visual evidence consistent with the theoretical expectation. It effectively demonstrates that democratic institutions serve as a powerful buffer against the fiscal pressures to repeal the inheritance tax, adding a crucial and intuitive nuance to the original authors' additive claims.

5 Conclusion

This replication study confirms the main findings of Genschel, Limberg, and Seelkopf (2024), demonstrating that the likelihood of inheritance tax repeal is strongly associated with changes in fiscal structure, particularly the expansion of more efficient revenue sources. The replicated results closely align with the original estimates, reinforcing the robustness and reproducibility of the authors' core arguments.

Building on this, the extension introduced in this paper highlights the potential moderating role of democratic institutions. While the interaction model suggests that democracies may attenuate the effect of revenue redundancy on tax repeal, the lack of statistical significance indicates that this relationship remains suggestive rather than conclusive.

Overall, this analysis illustrates both the strengths and limitations of the original model. While it provides a compelling explanation centered on fiscal capacity, further research could explore more complex interactions between political institutions and economic conditions. In particular, future work may benefit from addressing the rarity of repeal events and employing alternative estimation strategies to better capture such dynamics.

Taken together, this study demonstrates not only the empirical robustness of the original findings but also the value of extending replication exercises to generate new theoretical and methodological insights.